

Non-ideal compressible flow dynamics of two-phase supersonic expansions and their implications for nozzle design

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ABSTRACT

The growing interest in two-phase flows in turbomachinery has increased the need for systematic benchmarking of non-ideal effects under two-phase conditions. Under the homogeneous equilibrium assumption, the two-phase mixture behaves as a compressible fluid with a well-defined local speed of sound, giving rise to compressible flow phenomena similar to those governing single-phase gas dynamics. While non-ideal compressible flow dynamics have been extensively studied for single-phase gases, the corresponding behavior of compressible two-phase mixtures remains largely unexplored. To address this gap, we present an analysis of nonideal effects in representative two-phase isentropic expansions. These include flashing processes originating from subcooled conditions, as well as wet-to-dry expansions that start within the two-phase region and lead to complete fluid vaporization. The analysis is conducted by evaluating Mach number trends across different isentropes, assuming homogeneous equilibrium conditions between the phases. This analysis links two-phase behavior to the non-ideal effects known from single-phase gas dynamics, highlighting similarities in the underlying expansion mechanisms. On this basis, we present novel methods to account for non-ideal effects in two-phase flows when applying the method of characteristics to supersonic nozzle design. The results provide new insight into the sonic properties and fundamental expansion mechanisms of two-phase flows, and offer a practical framework for incorporating these effects into the preliminary design of supersonic nozzles.

Keywords: organic Rankine cycle, wet-to-dry cycle, flashing flows, method of characteristics, barotropic model

1 Introduction

Two-phase supersonic turbines have recently attracted interest for their applications in power generation, refrigeration, and liquefaction cycles. The need for advanced waste heat recovery has drawn attention to partial evaporation Organic Rankine Cycles (ORC), which can increase net power output by 20 % to 60 % [1]. In refrigeration and liquefaction cycles, two-phase expanders offer potential efficiency gains by replacing conventional Joule-Thomson valves, allowing the recovery of expansion work [2].

The low speed of sound characterizing two-phase flows, together with the high expansion ratios encountered in these systems, requires converging-diverging channels for the design of supersonic stator blades. The Method of Characteristics (MoC) represents a well-established technique for the design of supersonic nozzles and blades in ideal- and real-gas single-phase flows [3]. The MoC has recently been extended to two-phase flows under the Homogeneous Equilibrium Model (HEM) assumption for partial evaporation ORCs [4], achieving effective designs of two-phase stators operating with a two-phase mixture throughout the blades. However, when the expansion crosses the saturation dome, the HEM formulation of the speed of sound exhibits abrupt discontinuities which complicate the application of the MoC. Such discontinuities occur in wet-to-dry ORCs [5], where the fluid expands from saturated liquid or low quality to superheated vapor, crossing the vapor saturation line along the expansion. Additionally, in refrigeration and liquefaction systems, the expander inlet is typically at subcooled conditions, and the expansion crosses the two-phase dome at the liquid saturation line, where the speed of sound exhibits an even stronger discontinuity than in the wet-to-dry cases [6].

To address the lack of methods for applying the MoC across speed of sound discontinuities, we first characterize these discontinuities in relevant two-phase applications, focusing on: (i) how the resulting Mach number trends differ from the ideal-gas case, (ii) how different inlet conditions affect these non-ideal effects, and (iii) where the choking condition is

met along the isentropic expansions. Building on this physical characterization, we then propose strategies for handling discontinuities when using the MoC to design two-phase supersonic nozzles. The wet-to-dry discontinuity is handled by treating the resulting characteristic-line crossover with techniques developed to apply the MoC in the presence of weak oblique shock waves in single-phase flows [7](Sec. 16-4). On the other hand, the flashing case requires careful consideration at the throat, since the speed of sound discontinuity drives the flow from subsonic to highly supersonic conditions. The resulting nozzle geometries are then simulated using two-dimensional Computational Fluid Dynamics (CFD) simulations employing the barotropic flow model, to verify the extended MoC method and further investigate flow features of interest, such as the subsonic-to-supersonic transition and the flow field contours. To the best of the authors' knowledge, this represents the first extension of the MoC to two-phase nozzle design in the presence of speed of sound discontinuities, extending its applicability to expansions crossing the saturation lines.

2 Characterization of non-ideal effects in two-phase flows

This section presents the two case studies considered, along with a preliminary analysis of their non-ideal flow features. The first case concerns wet-to-dry expansions, starting from a two-phase mixture and ending in a superheated state at the nozzle outlet. The second case concerns flashing flows, starting from a subcooled state at the inlet and terminating in a two-phase mixture with relatively moderate vapor quality at the nozzle exit. In both cases, the fluid properties are calculated using the HEM assumption, with the speed of sound formulated as proposed by Flåtten and Lund [8]. Other relaxation formulations remain continuous but non-smooth across the saturation line, whereas the HEM one exhibits an abrupt discontinuity when crossing the two-phase dome [6].

2.1 Wet-to-dry expansions

For the wet-to-dry case, MM was selected as the working fluid, as it has been identified as an excellent candidate in terms of second-law efficiency for wet-to-dry cycles [5]. To draw an analogy between single-phase and two-phase non-ideal effects, both types of expansions are examined. Figure 1(a) shows the eight isentropic expansions considered, where the single-phase expansions start from $P_0 = 2P_{cr}$, with different total temperatures used to move the expansion closer to or further from the critical point, while the two-phase expansions start from the liquid saturation line. Figure 1(b) shows the single-phase Mach number trends against normalized pressure, where the increase in non-ideal effects can be noticed as the expansion approaches the two-phase dome. The Mach number departs from the perfect-gas behavior as the critical point is approached, becoming non-monotonic and showing a local maximum for the case with the lowest inlet temperature. This phenomenon is well known for single-phase flows and has been extensively studied in nozzles and nozzle cascades [9]. Figure 1(c) shows the Mach number against normalized pressure for the wet-to-dry cases considered. The expansion starting at $0.90 T_{cr}$ remains within the two-phase region throughout and does not exhibit any discontinuity in the Mach number trend, while the other expansions show a discontinuity where the isentrope crosses the vapor saturation line. By comparing the trends between single-phase and two-phase flows in Figs. 1(b) and 1(c), a clear analogy emerges between their non-monotonic behavior. In the single-phase case, the Mach number trend is continuous and exhibits a local maximum for the expansion closest to the critical point. In the two-phase case, the discontinuity at the phase-change boundary splits the trend into two piecewise monotonic segments. In both cases, the resulting curves are multi-valued: a single Mach number corresponds to multiple pressure values, whether due to the continuous maximum in the single-phase case or the discontinuous jump in the two-phase case. Finally, the Mach number trends for the wet-to-dry expansions also highlight the importance of an efficient design of the diverging section. Since sonic conditions are reached at a smaller expansion ratio compared to the perfect-gas case, most of the pressure drop occurs under supersonic conditions.

2.2 Flashing expansions

Nitrogen was considered as the working fluid for the flashing expansions, motivated by its relevance to air liquefaction and separation processes. Figure 2(a) shows the temperature–entropy diagram of nitrogen along the the four degrees of subcooling considered and their respective isentropic expansions, all starting from a stagnation pressure of 20 bar. The mass flux, shown in Fig. 2(b), exhibits a kink at the crossing point with the liquid saturation line, due to the sudden change in the slope of the density trend. The maximum of each curve represents the choking point of the respective expansion, where the Mach number is not necessarily equal to one, as it can instead jump directly from subsonic to well above unity. For all expansions except the one with the lowest subcooling, this maximum coincides with the kink, where the flow jumps directly from subsonic to supersonic conditions right at the intersection with the liquid saturation line. By contrast, for the case with 2.5 K of subcooling, the critical condition ($M = 1$) is reached slightly after crossing the saturation dome, and the kink does not coincide with the maximum. Figure 2(c) shows the Mach number as a function of normalized pressure for the four cases. Since the subcooled liquid is nearly incompressible, it features a high speed of sound, leading

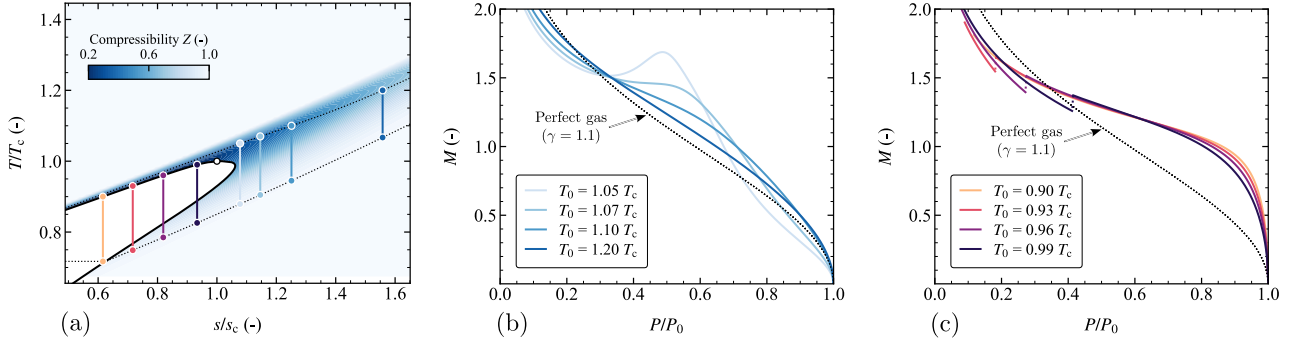


Figure 1: Preliminary study of wet-to-dry expansions. (a) T - s diagram of MM with the expansions considered; (b) Mach number as a function of normalized pressure for single-phase expansions in the non-ideal flow regime; (c) Mach number as a function of normalized pressure for wet-to-dry expansions.

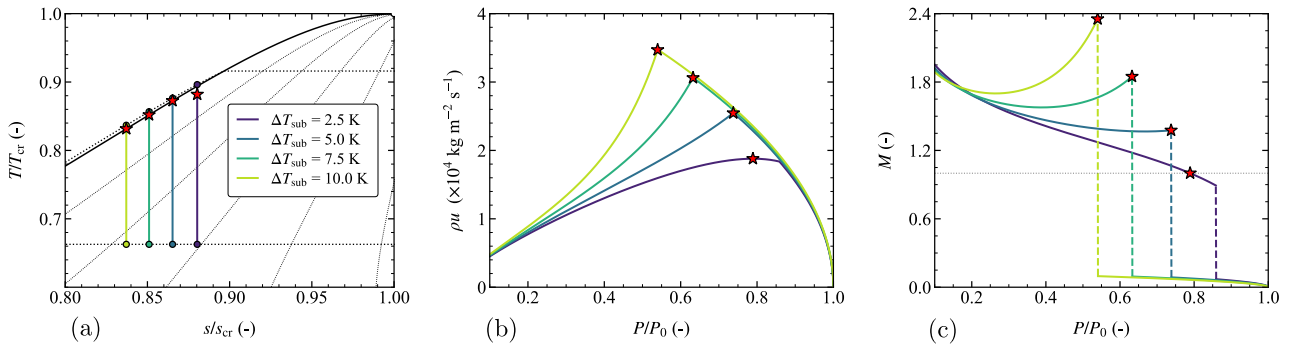


Figure 2: Preliminary study of flashing expansions. (a) T - s diagram of nitrogen showing the flashing expansions considered; (b) mass flux for the four cases; (c) Mach number as a function of normalized pressure. Red stars indicate where the critical condition is reached along each expansion.

to particularly low Mach numbers in the subcooled region of the expansion. Upon entering the two-phase region, an abrupt decrease in the speed of sound occurs, driving the Mach number to high values. For the 2.5 K subcooling case, the Mach number downstream of the discontinuity remains below one, and the sonic condition is met at the throat. The two-phase expansion then continues in the supersonic regime following a monotonic Mach number trend. For the other three cases, the Mach number after the discontinuity is well above the critical condition, reaching a value of 2.4 for the 10 K subcooling case. Additionally, for the 7.5 K and 10 K subcooling cases, the trend is non-monotonic, first decreasing and then increasing as the isentropic expansion proceeds from higher to lower pressures.

3 Extension of the Method of Characteristics to handle speed of sound discontinuities

The Mach number trends identified in Section 2 reveal where speed of sound discontinuities arise and how they affect the sonic transition, information that is essential for extending the MoC presented in Cioffi et al. [4] to design convergent-divergent nozzles for expansions that cross the saturation lines. Applying the MoC across a discontinuity requires the flow to remain supersonic. In wet-to-dry expansions, this requirement is inherently satisfied, since the flow crosses the vapor saturation line at a relatively high Mach number. The sonic transition is unaffected by any discontinuity, and the smooth passage through $M = 1$ allows a straightforward extension of the Sauer method [10] to predict the sonic line at the throat. The challenge that arises instead is the crossover of characteristic lines caused by the change in Mach angle, defined as $\sin^{-1}(1/M)$, as shown in Fig. 3(a). This effect, however, can be handled by allowing the characteristics to cross, in the same way as for weak oblique shock waves in single-phase flows [7]. The crossing generates a fold in the solution surface, which is fixed in the post-processing of the nozzle wall.

For flashing flows, the situation differs, as the discontinuity occurs near the throat, during the passage from subsonic to supersonic. For a low degree of subcooling, the sonic transition may still be smooth and occur within the two-phase region, as observed for the 2.5 K subcooling case in Fig. 2. In this case, the Sauer method can be applied in the same way as for the wet-to-dry expansions, as no discontinuities around $M = 1$ occur. To further assess the effect of using the Sauer line

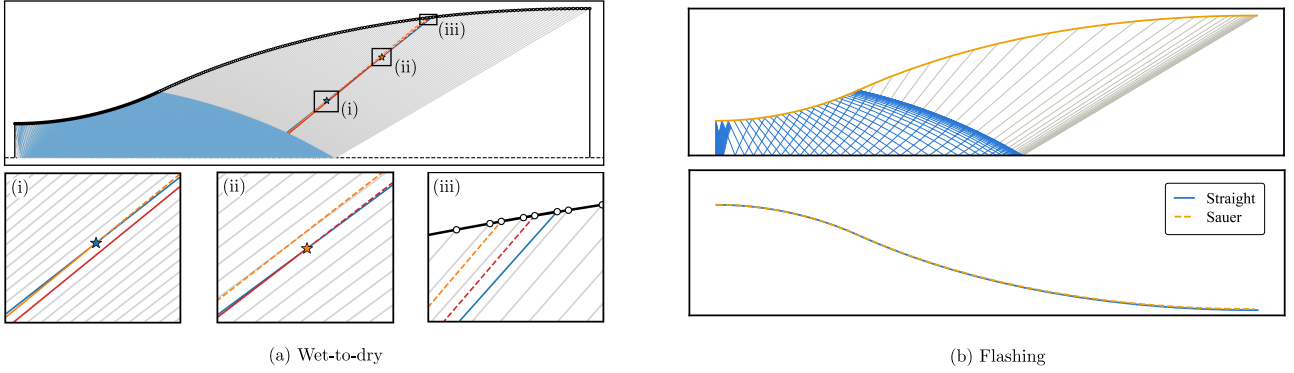


Figure 3: Characteristic grid for the MoC design of (a) wet-to-dry and (b) flashing expansions. In (a), three insets highlight the crossing characteristics when superheated conditions are reached. In (b), the bottom panel compares the nozzle shape obtained with a straight sonic line versus the Sauer parabolic one.

approximation, two nozzles were designed for the 2.5 K subcooling case: one using the Sauer line and one using a straight sonic line. The characteristic grid obtained with the straight sonic line is shown at the top of Fig. 3(b), while the geometries obtained with the Sauer line and the straight line are shown at the bottom of Fig. 3(b). No evident difference in nozzle shape is noticeable between the two approaches, indicating that for this case the nozzle geometry is largely insensitive to the specific choice of sonic line at the throat. For cases where the discontinuity occurs at the transition between the subsonic and supersonic regimes, the Sauer line is no longer applicable, since the small-perturbation technique is valid only for Mach numbers close to unity. Additionally, perturbing around the discontinuity would bring the flow back to subsonic conditions, where the Sauer analysis no longer holds. For these cases, a reasonable approximation is to assume a straight, vertical initial line, and the consequences of this assumption are further discussed in Sec. 4.2.

4 Results and discussion

This section presents the MoC predictions for the single-phase, wet-to-dry, and flashing cases and verifies them against two-dimensional CFD simulations. The CFD simulations are performed under the inviscid flow and HEM assumptions, consistent with the MoC framework. Specifically, a barotropic fluid model is implemented in Ansys Fluent through polynomial functions obtained by fitting data generated with CoolProp [11] via the barotropy package [12]. Inlet boundary conditions are prescribed via the total pressure and vapor quality, while the static pressure is imposed at the outlet.

4.1 Wet-to-dry expansion

The MoC has been applied to design convergent-divergent nozzles for two of the MM expansions analyzed in Section 2.1: the single-phase expansion starting from $T_0 = 1.05 T_{cr}$ and $P_0 = 2 P_{cr}$, and the wet-to-dry expansion from $P_0 = 4.78$ bar and $Q_0 = 0.3$. Figure 4 shows the comparison of Mach number and pressure distributions obtained from the MoC and the barotropic CFD model. The Mach number trend is smooth and non-monotonic, consistent with the near-critical non-ideal behavior discussed in Section 2. The MoC accurately reproduces this trend, with a maximum deviation below 1 % with respect to the CFD results.

Similarly, Figure 5 shows the same comparison for the wet-to-dry expansion. In this case, the Mach number trend is piecewise monotonically increasing, with a discontinuity at the vapor saturation line, where the flow enters the superheated vapor region shaded in Figs. 5(a) and (b). While the resulting Mach number contour in Fig. 5(c) resembles that of a weak oblique shock wave, the jump stems purely from the discontinuity in the speed of sound. The pressure field, by contrast, evolves smoothly toward the target outlet pressure. Both the Mach number and pressure trends agree closely between the MoC and the barotropic CFD model, with deviations below 1 %.

4.2 Flashing expansion

The MoC has also been applied to design a convergent-divergent nozzle for flashing flows. The exemplary case considered is the 10 K subcooling case of Fig. 2, as it exhibits the strongest non-ideal Mach number trend, due to the largest discontinuity and the pronounced non-monotonic behavior. Figure 6 shows the comparison between the MoC predictions and the CFD simulations for the (a) axial and (b) wall pressure and Mach number distributions, while (c) shows the Mach number and pressure contours obtained from post-processing of the CFD simulations. From the axial

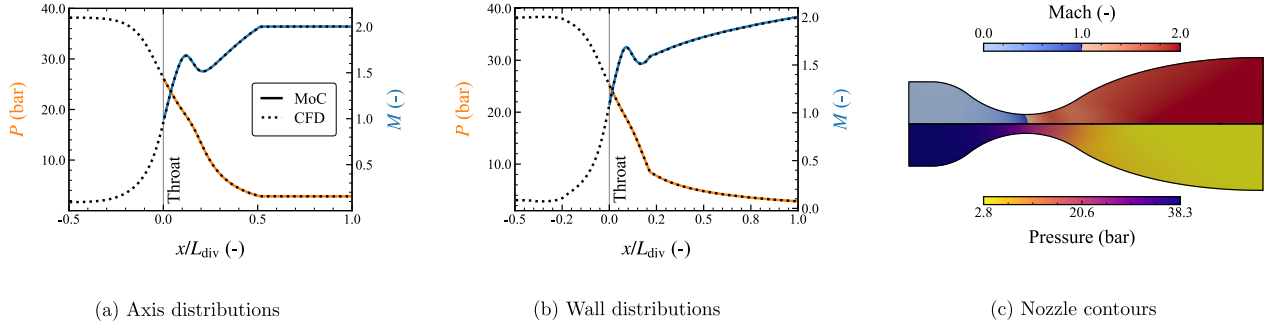


Figure 4: MoC verification against 2D inviscid CFD barotropic simulations for a single-phase non-ideal expansion. Mach number and pressure trends along the nozzle (a) axis and (b) wall; (c) CFD nozzle contours of Mach number and pressure.

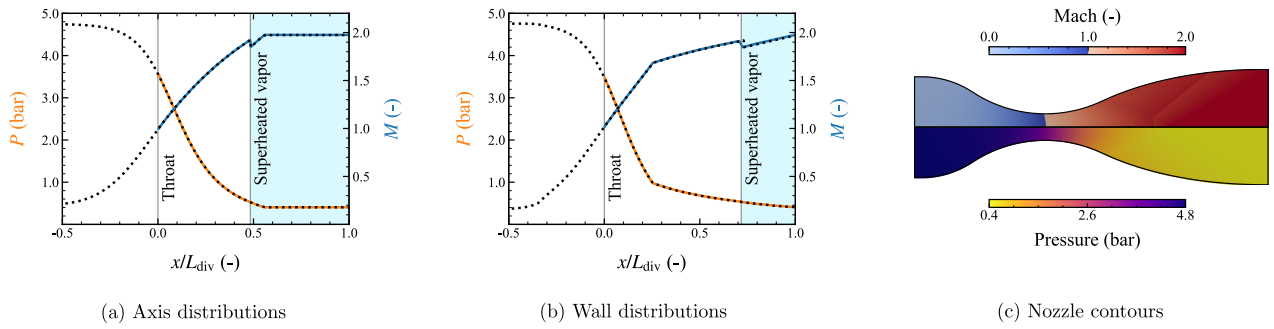


Figure 5: MoC verification against 2D inviscid CFD barotropic simulations for a wet-to-dry expansion. Continuous lines denote MoC results; dotted lines denote CFD results.

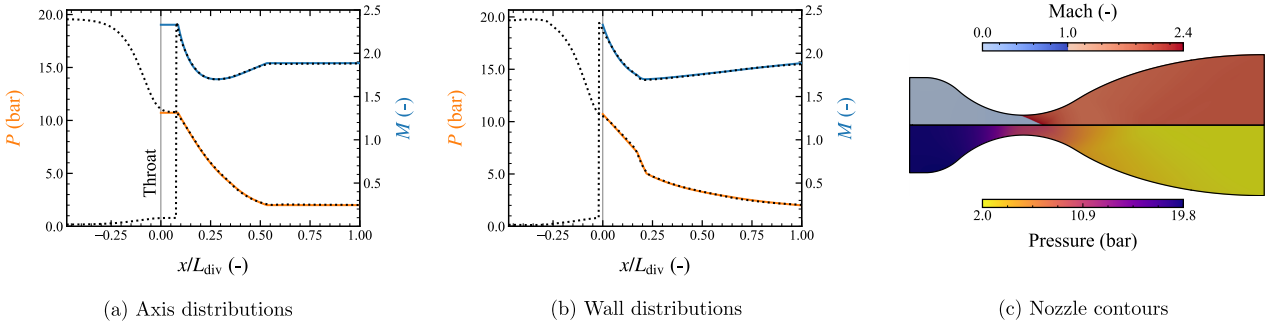


Figure 6: MoC verification against 2D inviscid CFD barotropic simulations for a flashing expansion. Continuous colored lines denote MoC results; dotted lines denote CFD results.

and wall trends, excellent agreement is achieved between the MoC and the CFD simulation, with a maximum error below 1 %, and the non-monotonous Mach number trend is well captured by the MoC. In the axial distributions of Fig. 6, the MoC trend near the throat remains constant over a short portion of the nozzle. This is a side effect of assuming a straight, vertical sonic line: due to how the initial value problem is set up in the MoC design procedure, the flow does not yet begin to expand in that region. However, this effect influences only a small portion of the nozzle and does not compromise the accuracy of the results downstream of the throat. In the Mach number contour of Fig. 6 obtained from the CFD simulation, an abrupt jump in Mach number can be noticed at the nozzle throat, and the sonic line no longer resembles a parabolic-like shape. Nonetheless, the pressure contour evolves smoothly toward the target outlet pressure, with no discontinuities present within the diverging section of the nozzle.

5 Conclusions and future works

This work characterized the speed of sound discontinuities arising in isentropic two-phase expansions under the Homogeneous Equilibrium Model (HEM) assumption, and extended the Method of Characteristics (MoC) to design

supersonic nozzles in their presence. The main findings can be summarized as follows:

- **Wet-to-dry expansions.** Across all cases considered, the Mach number trend is piecewise monotonic, with a single discontinuity occurring at the vapor saturation line and the flow remaining supersonic across it. This behavior allows the discontinuity to be treated as a characteristic-line crossover, analogous to a weak oblique shock wave in single-phase flows. The resulting extended MoC was verified against 2D inviscid CFD simulations using a barotropic fluid model, showing excellent agreement in the pressure and Mach number distributions along the nozzle axis and wall, with a maximum deviation below 1 %.
- **Flashing expansions.** The behavior of the discontinuity depends on the degree of inlet subcooling. At low subcooling, the flow remains subsonic after crossing the liquid saturation line and the subsequent sonic transition occurs within the two-phase region, following a smooth trend that allows the Sauer method to be applied without modification. At higher subcooling, the discontinuity coincides with the sonic transition itself, driving the flow directly from subsonic to supersonic conditions. Since the Sauer method relies on a small-perturbation analysis around the sonic point, it is no longer applicable in this regime, and a straight, vertical sonic line is instead assumed at the throat. The resulting extended MoC was verified against 2D inviscid CFD simulations using a barotropic fluid model, confirming that the method correctly captures the non-ideal Mach number trends predicted under the HEM assumption, with a maximum deviation below 1 %.

Overall, the proposed MoC provides a powerful preliminary design tool for two-phase nozzles in the presence of speed-of-sound discontinuities under the HEM formulation. Further work is needed to assess whether the resulting nozzle designs remain effective when non-equilibrium effects are considered in the performance assessment.

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